

✎ 침구의학 SUMMARY 공지사항

- 본시험: 말초신경 10개 선정하여 거기세어 60-80% 출제, 객관식 30문제로 CBT 형식으로 출제(증상을 묻고 포착된 말초신경을 묻거나 치료부위(혈자리)를 묻는 방식)
- 예비 시험: 5/31 금요일 2교시(10시~), CBT실, 수업 한시간 하고 2교시에 함께 이동하여 시행, 성적 반영 안하고 응시해야 출석으로 간주, 5문항 정도 나오며, 기말고사에 꼭 출제할 예정
- 혈자리를 시험에 내시겠다고 하셨는데.. 수업 시간에 혈 언급이 별로 없어서 아무래도 경혈들의 유주 분포를 암기해야 될 것으로 보입니다.

✎ 침구의학 SUMMARY 사용법

- 새로운 교수님으로 이전 학번 기출이 없습니다.
- 수업 내용이 깔끔하고, 어렵지 않기 때문에 공부해보시는 것을 추천드립니다.
- 중요해 보이는 내용은 가독성을 위해 **굵은 글씨**와 **빨간 글씨**로 표시하겠습니다.
- 사진이 너무나 많아서 적당한 분량을 위해 사이즈를 줄였습니다.

4절. 요둔부 포착신경 병증

수업 전 여담

- 명불허전 재밌다.

6. 후대퇴피부신경 포착 신경병증

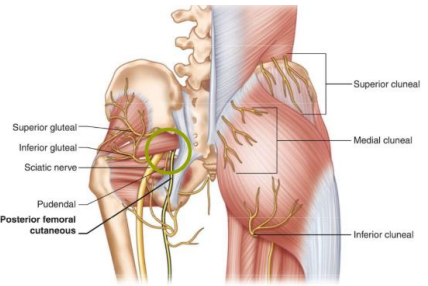
증상	• 둔부 하부에서 대퇴부 뒤쪽으로 퍼지는 통증 • 무릎 위쪽까지 통증이 발생할 수 있음 - 이상근 부위를 자극하면 다양한 피부 부위에 통증을 호소하는데, 이 중 하나가 대퇴 뒤쪽 부위의 후대퇴피부신경이다.
원인	• 둔부 주사 • 운동, 오래 앉아서 ischial tuberosity 부위 압박
포착 위치	• 이상근 하부 (foraminal infrapiriformis) • Ischial tuberosity  - 이상근 하부에서 나오기 때문에 해당 부위에서 포착이 일어날 수 있다.
통증 부위	- 증상호소는 대퇴 후면으로 말초신경이 지배하는 피부 영역의 통증을 호소한다.  Fig. 56.1 Patient pain complaint from posterior femoral cutaneous nerve entrapment (Image courtesy of Andrea Trescot, MD)

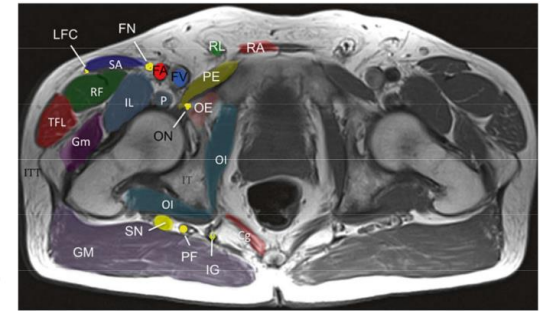


Fig. 56.2 Pain pattern from nerve entrapments of the posterior leg. *A* lateral branch iliohypogastric nerve, *B* superior cluneal nerve, *C* lateral femoral cutaneous nerve, *D* middle cluneal/sacral nerve, *E* inferior cluneal nerve, *F* posterior femoral cutaneous nerve, *G* obturator nerve, *H* femoral nerve, *I* saphenous nerve, *J* lateral sural cutaneous nerve, *K* superficial peroneal nerve, *L* medial calcaneal nerve (Image courtesy of Terry Dallas-Prunskis, MD)

- 대퇴 후면부 감각신경의 지배 영역으로 A: 장골하복신경 외측 가지, B: 상둔피신경, C: 외측대퇴표피신경, D: 중둔피신경, E: 하둔피신경, **F: 후대퇴표피신경**, G: 폐쇄신경, H: 대퇴신경, I: 복재신경, J: 비복신경, K: 표재종아리 신경 등
- 중둔피신경, 하둔피신경이 있고 바로 아래 후대퇴피부신경의 영역이 보인다.

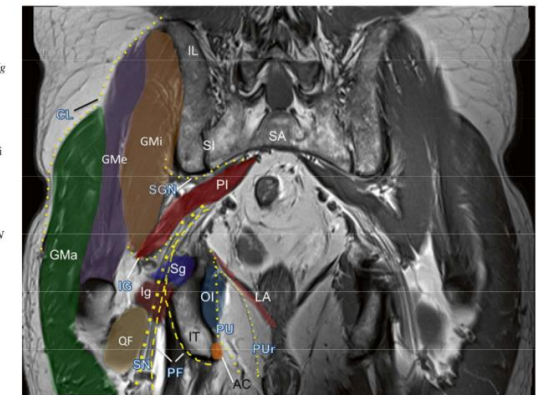
영상

Fig. 56.5 MRI axial image of the pelvis. *Cg* coccygeus muscle, *FA* femoral artery, *FN* femoral nerve, *FV* femoral vein, *GM* gluteus maximus muscle, *Gm* gluteus medius, *IG* inferior gluteal nerve, *IL* iliopectineus muscle, *IT* ischial tuberosity, *LFC* lateral femoral cutaneous nerve, *Of* obturator internus muscle, *P* psoas muscle, *PE* pectineus muscle, *PF* posterior femoral cutaneous nerve, *QF* quadratus femoris muscle, *RA* rectus abdominis muscle, *RF* rectus femoris muscle, *RL* round ligament, *SA* sartorius muscle, *SN* sciatic nerve, *TFN* tensor fascia lata muscle (Image courtesy of Andrea Trescot, MD)

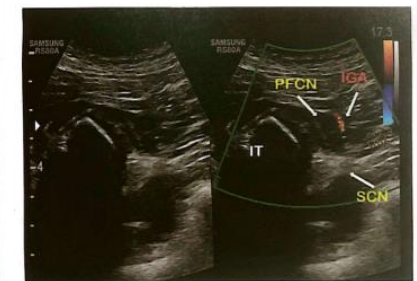
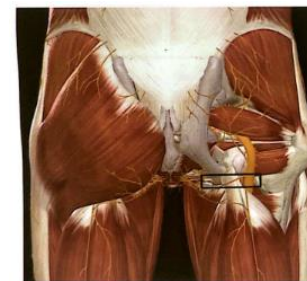


- 대퇴골 보이고, 위로 대둔근(GM) 있고, 바로 아래 좌골신경과 옆으로 후대퇴피부신경(PF)이 지나가는 것을 볼 수 있다.

Fig. 56.6 MRI coronal images showing gluteal muscles and nerves. *AC* Alcock's canal, *CL* superior cluneal nerve, *IL* iliac crest, *IG* inferior gluteal nerve, *Ig* inferior gemellus muscle, *IT* ischial tuberosity, *PI* piriformis, *PU* pudendal, *Gm* gluteus minimus, *Gme* gluteus medius, *Gma* gluteus maximus, *GT* greater trochanter, *LA* levator ani muscle, *Of* obturator internus muscle, *PF* posterior femoral cutaneous nerve, *PU* pudendal nerve, *PUr* pudendal nerve (rectal branch), *QF* quadratus femoris muscle, *SA* sacrum, *SGN* superior gluteal nerve, *Sg* superior gemellus muscle, *SI* sacroiliac joint, *SN* sciatic nerve (Image courtesy of Andrea Trescot, MD)



- 노란색 점선으로 후대퇴피부신경 확인(PF)



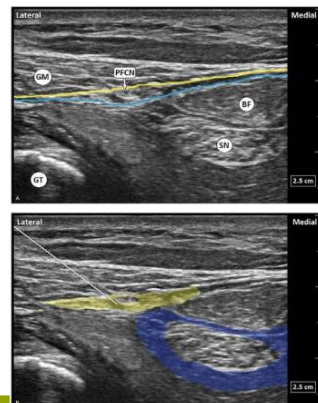
PFCN Posterior femoral cutaneous nerve **SCN** Sciatic nerve
IT Ishial tuberosity **IGA** gluteal artery

· 포착신경병증 - 후대퇴피부신경 포착신경병증

Figure 3. Injection approach.



image. BF indicates biceps femoris long head, GT, greater trochanter; GM, gluteus maximus; blue line, deep investing muscular fascia of the thigh; and yellow line, fascia lata. B. Simulated needle approach and latex deposition (blue around the SN and yellow around the PFCN).



추가 내용

- 블라인드 시술법으로 ABCD로 구분하여 접근하며, B를 기준으로 좌우에서 어디 부위에서 통증을 호소하는지 구분하여 치료

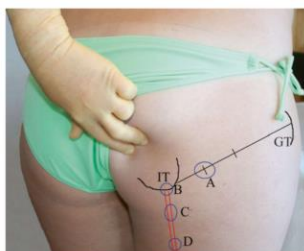
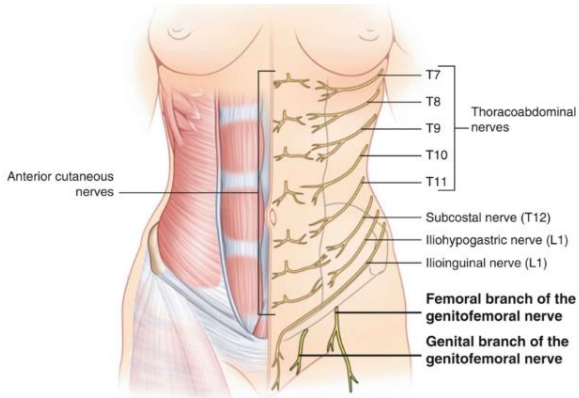


Fig. 56.9 Landmark-guided injections of the posterior femoral cutaneous nerve. IT ischial tuberosity, GT greater trochanter. A site described by Hughes [11], B injection at the ischium, C site recommended by Fritz [18], 2 cm below the IT, D site recommended by Tubbs et al. [11], 4 cm below the IT (Image courtesy of Andrea Trescot, MD).

- The site is in the gluteal fold, one quarter of the distance between the ischial tuberosity and the greater trochanter (Fig. 56.9 Site A). The authors suggest trying to feel two distinct losses of resistance as superficial and deep fascia are penetrated with a short-beveled needle.
- The perineal branch of the PFCN is usually injected at the ischial tuberosity (Fig. 56.9 Site B).
- Since the approach to the PFCN depends on the site of entrapment, Fritz et al. recommended injecting the nerve 2 cm below the ischial tuberosity (Fig. 56.9 Site C), proximal to the PBPFCN for patients with pain and paresthesia in the perineal area.
- Based on cadaver studies, Tubbs et al. suggested that the PBPFCN is on average located 4 cm inferior to the ischial tuberosity; they recommend injecting this area with anesthetic if this nerve is the source of the pain (Fig. 56.9 Site D).

7. 음부대퇴신경 포착신경병증

증상	• 서혜부, 회음부 통증, 감각장애 • 서혜부에서부터 회음부를 지나 대퇴 상부 가운데로 방 사되는 통증이나 감각장애 • 걷거나 대퇴 hyperextension시 악화 • 골반 수술 후 GFN neuropathy incidence: 0.3%
원인	• 골반 수술 • Psoas abscess, entrapment (lumbar plexopathy) • 임신(주로 말기) - 임신하면 골반강 내 공간이 좁아져서 여러 신경병증이 잘 발생한다. - 수술 후에 조직이 재생되면서 피부나 조직에 유착이 발생하면서 신경포착이 동반될 수 있다.
포착 위치	• 대요근 • Pubic tubercle • 수술부위(수술 부위 조직 재생으로 수년 후에 나타나 기도함) 

통증 부위



Fig. 45.1 Patient complaints of genitofemoral pain (Image courtesy of Andrea Trescot, MD)



Fig. 45.2 Pain pattern from genitofemoral neuralgia (Image courtesy of Andrea Trescot, MD)

- 대퇴전면부, 회음부 방향으로 통증을 호소한다.

영상

- 골반쪽의 치골 부위를 촉진하여 기준으로 하여 통증 유발 부위를 치료한다.



Fig. 45.10 Location of ultrasound probe for genitofemoral entrapment (Image courtesy of Andrea Trescot, MD)

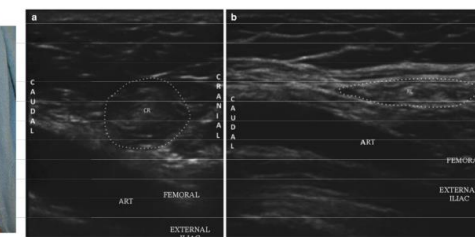


Fig. 45.11 Composite cross-sectional ultrasound image of the internal inguinal ring. Dashed line represents the internal inguinal ring; (a) male, (b) female. CR cremaster, RL round ligament, ART femoral artery (superficial) and external iliac artery (deep) (Image courtesy of Thiago Noser Frederico, MD)

- 초음파 사진 왼쪽은 남자, 오른쪽은 여성으로 여성의 경우, 난소와 자궁 연결하는 부위에 신경이 지나가는 것을 확인할 수 있다.



Fig. 45.10 Location of ultrasound probe for genitofemoral entrapment (Image courtesy of Andrea Trescot, MD)

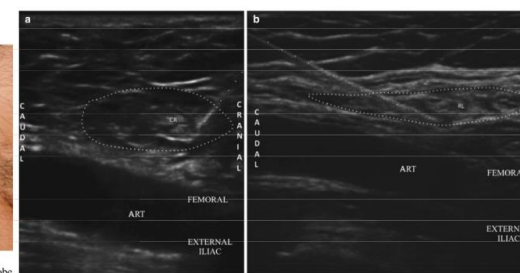


Fig. 45.12 Cross-sectional ultrasound image of the internal inguinal ring filled with local anesthetic and needle tip near the cremaster or round ligament. Dotted line represents the internal inguinal ring; (a) male, (b) female. CR cremaster, RL round ligament, ART femoral artery (superficial) and external iliac artery (deep) (Image courtesy of Thiago Noser Frederico, MD)

- 남성은 Cremaster(고환거근, CR)을 확인할 수 있으며, 여성은 Round ligament(RL)을 확인할 수 있다.

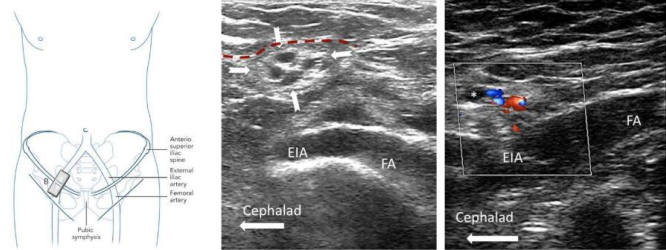


Figure 3a. Long axis view of the femoral artery and the external iliac artery showing the spermatic cord in cross section (outlined by solid arrows) in a male patient. The red dashed line outlines the deep abdominal fascia. Reproduced with permission from USRA (www.usra.ca)

- 사진에서 보는 것과 같이 혈액이 지나가는 부위에 신경도 함께 지나가므로 이를 고려하여 관찰한다.

추가 내용



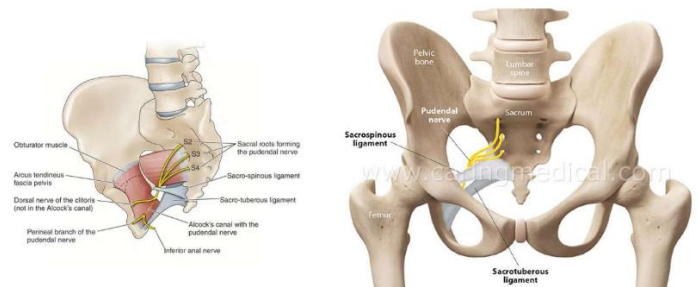
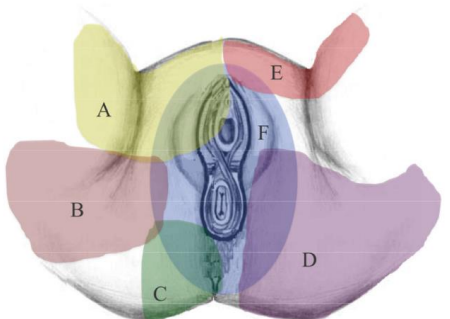
Fig. 45.10 Location of ultrasound probe for genitofemoral identification (Image courtesy of Andrea Trescot, MD)

- Trescot described a landmark-guided ("blind") technique to inject the genital branch of the GFN. The patient is placed in the supine position with a pillow under the knees if extending the lower limbs evokes pain. After a sterile skin prep, the pubic tubercle is palpated, and 1 cc of local anesthetic and deposteroid is injected via a 25-gauge needle, just superior and lateral to the tubercle.



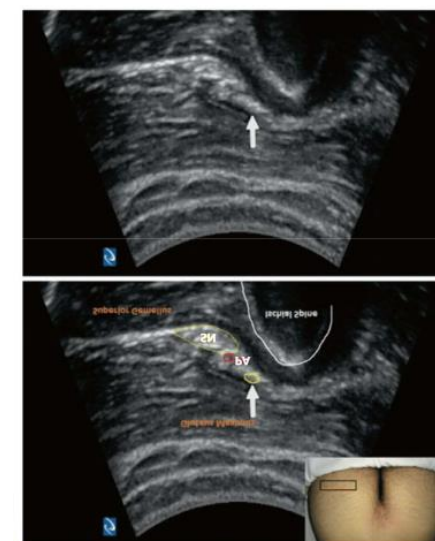
- 블라인드 시술법으로 무릎 아래 베개를 놓고 양와위로 누운 다음, 치골 결절을 만져 바로 위쪽 측면을 목표로 시술한다.

8. 음부신경 포착신경병증

증상	• 생식기, 항문직장 주변의 통증 • 생식기나 직장의 이물감, 감각장애 • 앉아있을 때 악화 • Incidence rate 1:100,000 • Alcock's syndrome (pudendal neuralgia) - CTS와 유사하다 (Angeles 2005)
원인	• 수술/출산 등 • 자전거 라이딩, 오래 앉아있는 경우
포착 위치	• 이상근 • Ischial spine • Sacrotuberous ligament • Alcock's canal  - 이상근 기시점에서 함께 나오기 때문에 이상근에 의해 포착
통증 부위	Fig. 47.1 Innervation of the perineum: A genitofemoral nerve, B obturator nerve, C inferior cluneal nerve, D peroneal branch of the posterior femoral cutaneous nerve, E ilioinguinal nerve, and F pudendal nerve (Image inspired by Hibner et al. [5], courtesy of Andrea Trescot, MD)  A: 음부대퇴신경, B: 폐쇄신경, C: 하둔피신경, D: 후대퇴피부신경, E: 엉덩고살신경, F: 음부신경

영상

Figure 2 (continued) MD
A: Pudendal nerve; B: Iliac vessels; C: Iliac vessels; D: Iliac vessels; E: Iliac vessels; F: Iliac vessels; G: Iliac vessels; H: Iliac vessels; I: Iliac vessels; J: Iliac vessels; K: Iliac vessels; L: Iliac vessels; M: Iliac vessels; N: Iliac vessels; O: Iliac vessels; P: Iliac vessels; Q: Iliac vessels; R: Iliac vessels; S: Iliac vessels; T: Iliac vessels; U: Iliac vessels; V: Iliac vessels; W: Iliac vessels; X: Iliac vessels; Y: Iliac vessels; Z: Iliac vessels; AA: Iliac vessels; AB: Iliac vessels; AC: Iliac vessels; AD: Iliac vessels; AE: Iliac vessels; AF: Iliac vessels; AG: Iliac vessels; AH: Iliac vessels; AI: Iliac vessels; AJ: Iliac vessels; AK: Iliac vessels; AL: Iliac vessels; AM: Iliac vessels; AN: Iliac vessels; AO: Iliac vessels; AP: Iliac vessels; AQ: Iliac vessels; AR: Iliac vessels; AS: Iliac vessels; AT: Iliac vessels; AU: Iliac vessels; AV: Iliac vessels; AW: Iliac vessels; AX: Iliac vessels; AY: Iliac vessels; AZ: Iliac vessels; BA: Iliac vessels; BB: Iliac vessels; 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- 사진과 같이 프로브를 대고 관찰하면 좌골을 확인하고, 도플러를 사용하여 음부 동맥을 찾아 바로 옆 부위의 신경을 확인한다.

추가 내용

Fig. 47.1 Innervation of the perineum: A genitofemoral nerve, B obturator nerve, C inferior cluneal nerve, D peroneal branch of the posterior femoral cutaneous nerve, E ilioinguinal nerve, and F pudendal nerve (Image inspired by Hibner et al. [5], courtesy of Andrea Trescot, MD)

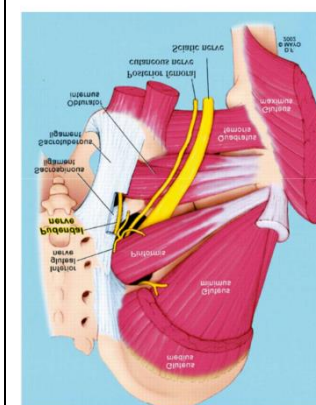
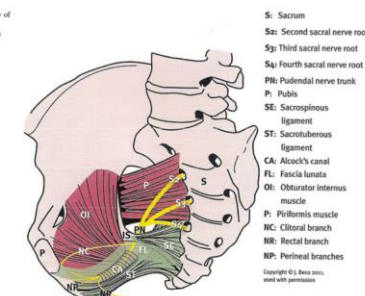


Fig. 47.4 Schematic anatomy of the pudendal canal (2011) Jacques Beau M.D., used with permission



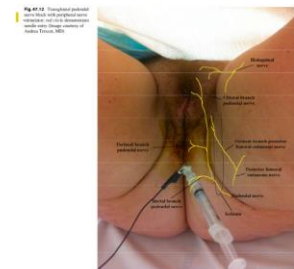
- 그림으로 신경 주행 확인



- Intravaginal approach
- Transrectal approach
- To perform a left-sided block, palpate the **ischial spine** with the index finger of the left hand, hold the syringe in the right hand, and guide the needle between the index and middle fingers of the left hand toward the ischial spine (Fig. 47.11).



Fig. 47.11 Landmark-guided technique – intrapelvic pudendal nerve injection (Image inspired by Prendergast et al. [15], courtesy of Springer)



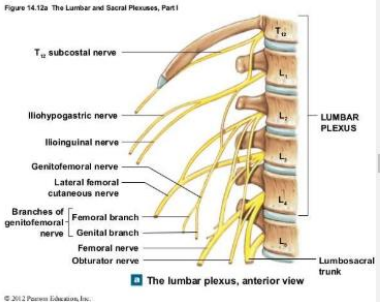
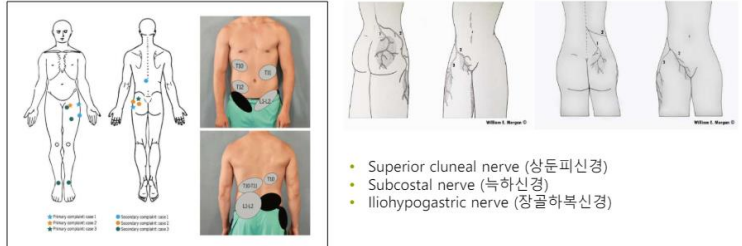
- Transgluteal approach



- Alcock's canal

- 블라인드 시술법으로 회음부 안쪽이나 항문 안쪽으로 들어가서 접근할 수 있으며, 둔부 아래쪽에서 접근하는 방법도 있다.

9. 흉요추 연접부 증후군

증상	• T-L spine 연접부(T10-L2) 신경 압박이 발생할 때 발생하는 일련의 증상들의 통칭 • 요추신경 후외측지 압박 증후군이라고도 함 • 요통, 골반에서 하지로 방사되는 통증 • 복부 통증이나 내장 증상도 동반할 수 있음 • Maigne's syndrome
원인	• 대요근 긴장 • 관절 기능 부전
포착 위치	• Lumbar plexus 주변 • 후관절 등  Figure 14.12a: The Lumbar and Sacral Plexuses, Part I T₁₂ subcostal nerve Iliohypogastric nerve Ilioinguinal nerve Genitofemoral nerve Lateral femoral cutaneous nerve Branches of genitofemoral nerve Femoral branch Femoral nerve Obturator nerve LUMBAR PLEXUS Lumbosacral trunk The lumbar plexus, anterior view © 2012 Pearson Education, Inc. - T10-L2에서 나오는 신경들인 상둔피신경, 늑하신경, 장골하복신경 등의 압박으로 인한 증상들의 통칭이다.
통증 부위	- 허리 쪽으로 알 수 없는 통증 호소  • Superior cluneal nerve (상둔피신경) • Subcostal nerve (늑하신경) • Iliohypogastric nerve (장골하복신경)

추가 내용

Clinical Case Report

Medicine

OPEN

Acupuncture for patients with Maigne's syndrome

A case series

Heesol Lee, KMD, PhD^a, Hyocheong Chae, KMD, BA^a, Myungseok Ryu, KMD, BA^a, Changsoo Yang, KMD, PhD^a, Sungba Kim, KMD, PhD^{a,*}

Abstract

Rationale: Diagnosing the precise etiology of low back pain (LBP) is crucial for facilitating speedy recovery in patients; Maigne's syndrome (MS), commonly referred to as thoracolumbar junction syndrome, is a condition characterized by pain resulting from nerve entrapment, yet its underlying mechanisms remain poorly understood. This study presents a series of six case reports wherein patients diagnosed with MS received acupuncture treatment.

- 흉요추 연접부 증후군 침 치료 논문으로, 아래 표와 그림과 같이 치료 포인트를 잡을 수 있다.

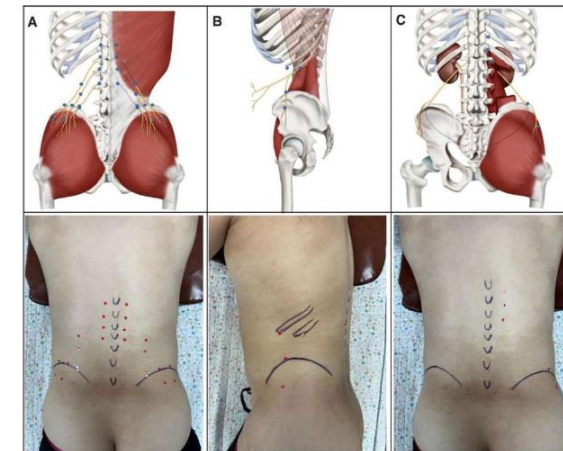


Figure 2. Acupoints of entrapment of nerves. (A) Acupoints of the superior cluneal nerve (SCN); (B) acupoints of subcostal nerve; and (C) acupoints of the iliohypogastric nerve.

Table 2

Characteristics of cases.

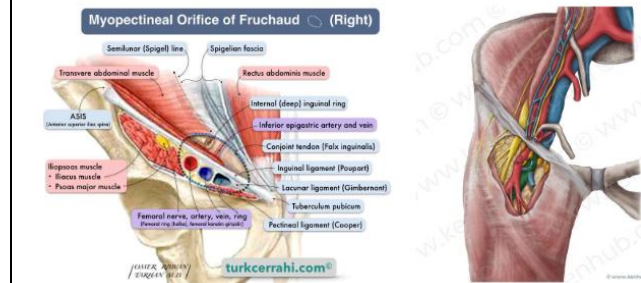
	Case 1	Case 2	Case 3	Case 4	Case 5	Case 6
Age, sex	45, Male	74, Female	55, Female	52, Male	54, Female	48, Male
Height (cm)/Weight (kg)	171/90	153/50	162/65	170/79	154/58	174/78
Chief complaint	LBP bilateral, both hips and inguinal areas pain	LBP bilateral, belt distribution pain	LBP bilateral, pain when standing up to work	Left LBP and left flank pain	LBP difficulty moving or turning	LBP bilateral, worse over the weekend
Diagnosis	MS, entrapment of SCN, SN, IN	MS, entrapment of SCN	MS, entrapment of SCN, IN	MS, entrapment of SCN, IN	MS, entrapment of SCN, IN	MS, entrapment of SCN, IN
Lumbar flexion test	+	ND	+	ND	+	+
Tibial nerve compression test	+	+	+	+	+	+
Pinch-tell test	Region of SCN, SN, IN	Region of SCN	Region of SCN & IN	Region of SCN & IN	Region of SCN	Region of SCN & IN
Thoracic vertebrae compression test	+	+	+	+	+	+
Past history	Urethral cancer, left kidney organ donation		Facial paralysis, cervical radiculopathy			
Onset	2020-04-25	2020-02-02	2021-02-24	2021-02-02	2021-01-23	2021-03-13
First treatment date	2021-01-25	2021-02-27	2021-02-27	2021-02-27	2021-02-25	2021-03-19
Treatment period	3 times for 8 d	2 times for 3 d + 50 d (0/4)	2 times for 8 d + 22 d (0/4)	3 times for 8 d	3 times for 8 d + 134 d (0/4)	2 times for 8 d
Interventions	Acupuncture, cupping, ICT, HP, HM	Acu. cupping, ICT, HP	Acu. cupping, ICT, HP	Acu. cupping, ICT, HP	Acu. cupping, ICT, HP	Acupuncture, cupping, ICT, HP, HM

Acu = acupuncture, Iv = follow up, HM = herbal medicine, HP = hot pack, ICT = interferential current therapy, IN = iliohypogastric nerve, LBP = low back pain, MS = Maigne's syndrome, ND = no data, SCN = superior cluneal nerve, SN = subcostal nerve.

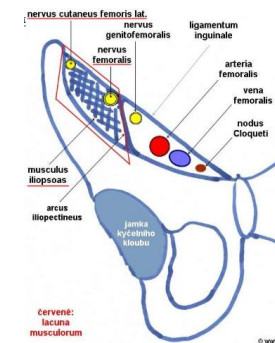
5절. 하지부 포착신경 병증

1. 대퇴신경 포착신경병증

증상	• 대퇴 전면의 가장 큰 신경 • 서혜부, 대퇴전면부, 종아리 중앙부, 발, 엄지의 저림 통증 유발 • 무릎 위약감이나 잦은 넘어짐 유발 가능 • 심할 경우 대퇴사두근 위약
원인	• 골반, 대퇴골 인공관절수술(수술 후 발생률 0.1-2.3%) • 장요근 혈종, 동맥류 등 • 오랫동안 대퇴를 구부리는 자세 (스키, 쪼그려 앉는 자세)
포착 위치	• Iliac compartment • Inguinal ligament • Lacuna musculorum Fig. 57.4 Anatomy of the femoral and saphenous nerve (Image courtesy of Springer)



- 그림과 같이 서혜인대 안쪽으로 신경이 주행하기 때문에 서혜인대에서 포착될 수 있다.



- Lacuna musculorum (muscular lacuna)
- The muscular lacuna (Latin: lacuna musculorum) is the lateral compartment of the thigh **beneath the inguinal ligament**. It is separated from the medial vascular lacuna by the iliopectineal arch.

- 위쪽 부위는 신경이 위주로 주행하며, 아래쪽은 혈관들이 주행한다.

통증 부위



Fig. 57.1 Patient identification of femoral nerve pain (Image courtesy of Andrea Trescott, MD)



Fig. 57.2 Pattern of pain from anterior lower extremity nerve entrapment. A: inguinal nerve, B: lateral femoral cutaneous nerve, C: lateral femoral cutaneous nerve, D: obturator, E: lateral femoral cutaneous nerve, F: saphenous nerve, G: superficial peroneal nerve, H: deep peroneal nerve, I: sural nerve, J: plantar nerve (Image courtesy of Andrea Trescott, MD)

- 대퇴전면부 방향으로 통증을 호소하며, 빨간색 B 부위가 지배 영역이다.(다른 신경 지배 영역도 이후에 계속 배울 것이다)
- A: 음부대퇴신경, **B: 대퇴신경**, C: 외대퇴피부신경, D: 폐쇄신경, E: 외측 비복대퇴신경, F: 복재신경, G: 표재 종아리신경, H: 심부 종아리신경, I: 비복신경

영상



Fig. 57.12 Ultrasound out-of-plane injection of the femoral nerve, showing probe and needle position (Image courtesy of Andrea Trescott, MD)

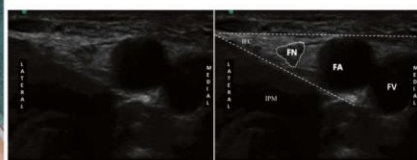


Fig. 57.13 Ultrasound image of the femoral nerve. FN: femoral nerve, FA: femoral artery, IPM: inferior psoas muscle, IFC: inferior femoral cutaneous nerve (Image courtesy of Thiago Nouer Frederico, MD)

- 서혜부에 프로브를 대어 대퇴 동맥을 확인하고, 신경 주행을 확인한다.



Fig. 57.13 Ultrasound in-plane injection of the femoral nerve, showing probe and needle position (Image courtesy of Thiago Nouer Frederico, MD)

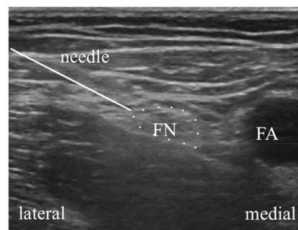


Fig. 57.14 Ultrasound image of simulated needle placement for femoral nerve injection (Image courtesy of Andrea Trescott, MD)

추가 내용

CASE REPORT

Delayed iliacus compartment syndrome following femoral artery puncture: case report and literature review

Bibombe P. Mwijipatayi^{1,2,*}, Ali Daneshmand¹, Haider K. Bangash¹, and Jackie Wong²

¹Department of Vascular Surgery, Royal Perth Hospital, Perth, Australia, and ²School of Surgery, Faculty of Medicine, Dentistry and Health Sciences, University of Western Australia, Perth WA 6009, Australia

*Correspondence address: Department of Vascular Surgery, Royal Perth Hospital, Perth, Australia and School of Surgery, Faculty of Medicine, Dentistry and Health Sciences, University of Western Australia, Level 2, 3441 Paddington Ave, W Perth WA 6150, Australia. Tel: +61 8 9276 2000; Fax: +61 8 9276 2000; E-mail: bibombe.p@perth.hosp.wa.gov.au; bibombe@perth.wa.gov.au

INTRODUCTION

Iliacus compartment syndrome (ICS) is a rare retroperitoneal compartment neuropathy caused by bleeding within the iliacus muscle leading to hematoma formation and compression upon the femoral nerve (FN). Due to the subsequent ischaemia of the femoral nerve, the symptomatology of ICS follows the innervation distribution of the femoral nerve causing both sensory and motor deficits.



Figure 1: (a) and (b) Axial and coronal computed tomography scans demonstrating a large iliacus hematoma in the right iliac fossa

- femoral artery은 큰 부위로 혈액 검사시에 활용하기도 하지만 쉽게 지혈도 잘 되지 않고, 주변의 구조물을 압박할 수 있다.
- MRI 사진 보면 해당 동맥이 손상된 것이 보이며, 서혜부 자침 할 때도 조심해야 한다.



Fig. 57.9 Landmark-guided injection of the femoral nerve (Image courtesy of Andrea Trescott, MD)

- The patient is placed in the supine position. The ipsilateral leg is abducted 10–20°, and the femoral artery palpated at the level of the femoral crease, below the inguinal ligament. The needle insertion site is approximately 1 cm lateral to the palpated pulse (Fig. 57.9).
- The needle is advanced at a 45° angle to the thigh in a cephalad direction. The clinician may notice a discreet change in resistance as the needle pierces the fascia lata, and the patient may report a paresthesia as the needle tip approaches the FN.

- 블라인드 시술법으로 대퇴 동맥을 잡고, 바깥쪽으로 1cm 정도 이동하여 치료 포인트로 한다.

수업 전 여담

- 교수님께서 인상깊게 읽으신 논문으로 여러 진통제와 침 치료의 치료효과를 비교하는 연구이다.
- 연구 결과로 침과 약물 치료의 치료효과가 큰 차이가 나지 않는 것을 확인할 수 있다. (응급상황에 침 치료와 약물 치료가 둘 다 효과가 없다 or 침 치료와 약물 치료 효과가 비슷하다)

(관심있으신 분은 아래 논문 읽어보세요)

Acupuncture for analgesia in the emergency department: a multicentre, randomised, equivalence and non-inferiority trial

Marc M Cohen ¹, De Villiers Smit ², Nick Andrianopoulos ³, Michael Ben-Meir ⁴, David McD Taylor ⁵, Shefton J Parker ⁶, Chalie C Xue ⁶, Peter A Cameron ²

Affiliations + expand

PMID: 28918732 DOI: 10.5694/mja16.00771

Free article

Abstract

Objectives: This study aimed to assess analgesia provided by acupuncture, alone or in combination with pharmacotherapy, to patients presenting to emergency departments with acute low back pain, migraine or ankle sprain.

Design: A pragmatic, multicentre, randomised, assessor-blinded, equivalence and non-inferiority trial of analgesia, comparing acupuncture alone, acupuncture plus pharmacotherapy, and pharmacotherapy alone for alleviating pain in the emergency department. Setting, participants: Patients presenting to emergency departments in one of four tertiary hospitals in Melbourne with acute low back pain, migraine, or ankle sprain, and with a pain score on a 10-point verbal numerical rating scale (VNRS) of at least 4.

Main outcome measures: The primary outcome measure was pain at one hour (T1). Clinically relevant pain relief was defined as achieving a VNRS score below 4, and statistically relevant pain relief as a reduction in VNRS score of greater than 2 units.

Results: 1964 patients were assessed between January 2010 and December 2011; 528 patients with acute low back pain (270 patients), migraine (92) or ankle sprain (166) were randomised to acupuncture alone (177 patients), acupuncture plus pharmacotherapy (178) or pharmacotherapy alone (173). Equivalence and non-inferiority of treatment groups was found overall and for the low back pain and ankle sprain groups in both intention-to-treat and per protocol (PP) analyses, except in the PP equivalence testing of the ankle sprain group. 15.6% of patients had clinically relevant pain relief and 36.9% had statistically relevant pain relief at T1; there were no between-group differences.

Conclusion: The effectiveness of acupuncture in providing acute analgesia for patients with back pain and ankle sprain was comparable with that of pharmacotherapy. Acupuncture is a safe and acceptable form of analgesia, but none of the examined therapies provided optimal acute analgesia. More effective options are needed.

Trial registration: Australian New Zealand Clinical Trials Registry, ACTRN12609000989246.

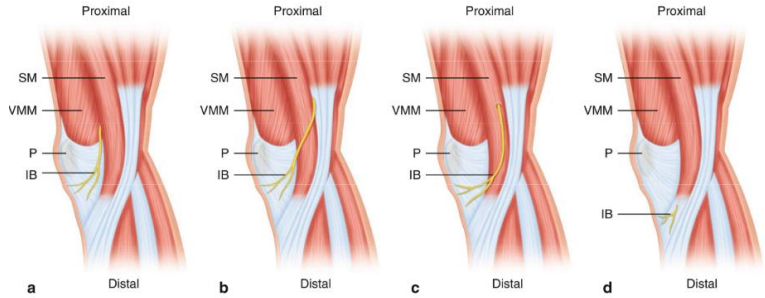
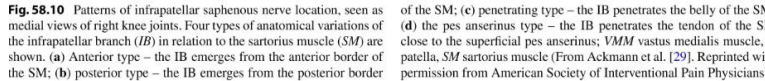
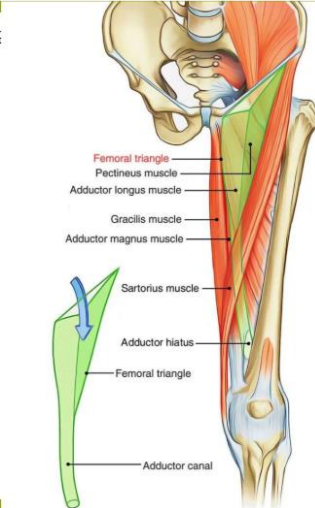
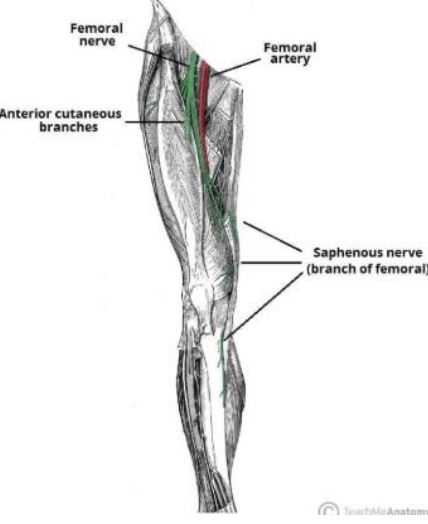
영상 검사 케이스 사이트

- <https://radiopaedia.org/>
- 교수님께서 애용하는 사이트로, 초음파, MRI, X-ray 까지 질환별로 여러 검사 사진들이 있다.

(관심있으신 분들은 들어가보시길)

The screenshot shows the Radiopaedia website. The header is dark blue with the Radiopaedia logo on the left and 'LOG IN' and 'SIGN UP' buttons on the right. Below the header, there are navigation links for 'ARTICLES', 'CASES', 'COURSES', and 'QUIZ'. A search bar is located on the right side of the header. Below the header, there is a banner for 'Radiopaedia 2024' conference. The main content area features a 'Case of the Day' section with a chest X-ray and text about 'Right middle lobe pneumonia with silhouetting of the right heart border'. To the right of the main content, there is a sidebar with statistics: '58688 cases' and '16879 articles'. Below the statistics, there is an advertisement for Adobe Creative Cloud.

2. 복재신경 포착신경병증(근위부)

증상	• 사타구니, 대퇴 중간부 내측의 통증 및 감각장애 • 무릎 내측, 엄지발가락으로도 방사되는 통증 및 감각장애
원인	• 대퇴부 혈관 관련 시술 (cardiac catheterization, femoral vasulcar surgery 등) • 서핑 등 운동 - 서핑과 같이 허벅지 쪽 근육을 많이 사용하는 경우에 포착될 수 있다.
포착 위치	• Inguinal ligament • Proximal thigh • Adductor canal • Infrapatellar saphenous nerve     • Adductor canal: (후내측)장내전근, 대내전근/ (전외측)내측광근/ (전내측)천공근막이 이루는 공간으로 복재신경이 지나가 포착이 나타날 수 있다.

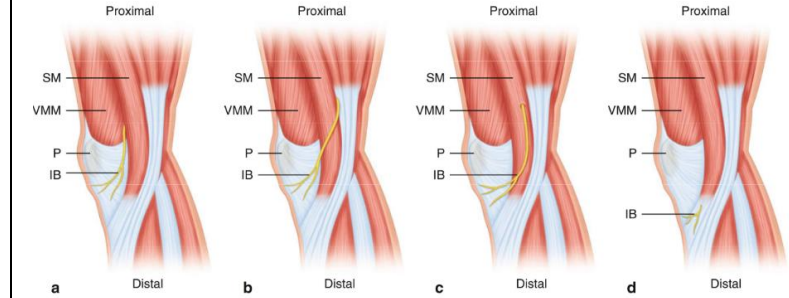


Fig. 58.10 Patterns of infrapatellar saphenous nerve location, seen as medial views of right knee joints. Four types of anatomical variations of the infrapatellar branch (IB) in relation to the sartorius muscle (SM) are shown. (a) Anterior type - the IB emerges from the anterior border of the SM; (b) posterior type - the IB emerges from the posterior border of the SM; (c) penetrating type - the IB penetrates the belly of the SM; (d) the IB penetrates the tendon of the SM close to the superficial pes anserinus; VMM vastus medialis muscle, P patella, SM sartorius muscle (From Ackmann et al. [29]. Reprinted with permission from American Society of Interventional Pain Physicians)

- 환자마다 복재신경의 주행 경로가 약간씩 다를 수 있으며, 핀 치를 검사를 통해 이를 구분하는 방법이 있다.

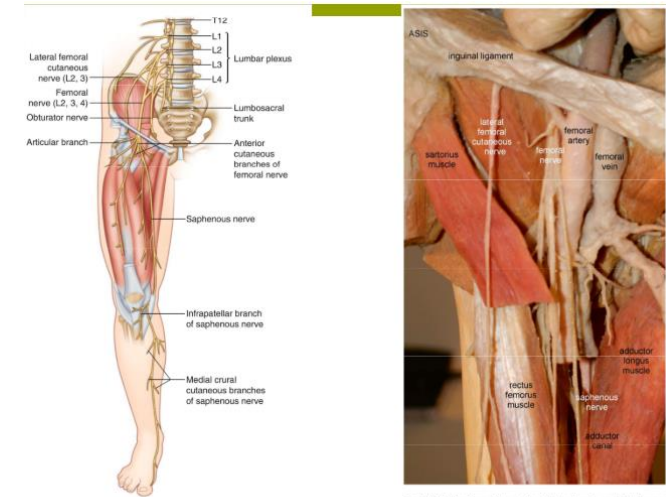


Fig. 58.5 Anatomy of the anterior thigh (Image from Springer)

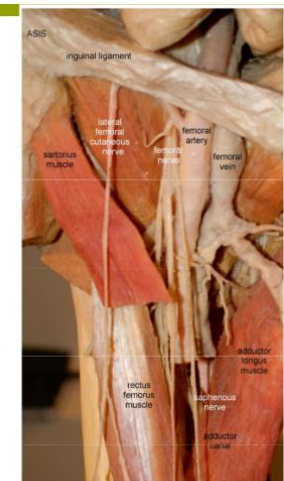


Fig. 58.6 Dissection of the anterior thigh and groin, modified from an image from Bodily, The Exhibition, with permission (Image courtesy of Andrea Trescot, MD)

통증 부위



Fig. 58.1 Patient pain complaints from proximal saphenous nerve entrapment (Image courtesy of Andrea Trescot, MD)



Fig. 58.2 Pattern of pain from proximal saphenous nerve entrapment (Image courtesy of Andrea Trescot, MD)

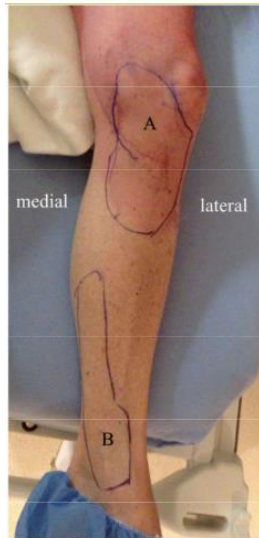


Fig. 58.3 Pattern of pain in a patient with proximal saphenous entrapment (infrapatellar saphenous) (a) and distal saphenous nerve entrapment (b) (Image courtesy of Agnes Szigöcs, MD)



Fig. 58.4 Tenderness over the adductor canal (Image courtesy of Andrea Trescot, MD)

- 해당 사진은 근위부, 원위부를 포함하여 나타낸 사진이며, 하지 내측에서 통증을 주로 호소한다.

영상

프로브를 무릎 아래 쪽으로 대어 복재신경이 나오는 것을 확인할 수 있다.

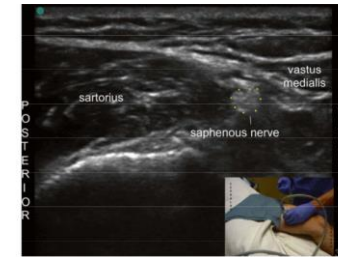


Fig. 58.19 Ultrasound identification and injection of the proximal saphenous nerve in the adductor canal (Image courtesy of Thiago Nasser Fernandes, MD)



Fig. 58.21 Ultrasound identification of the infrapatellar saphenous nerve. (a) Ultrasound probe location for infrapatellar nerve identification; (b) ultrasound image of the infrapatellar saphenous nerve; nerve is circled (Image courtesy of Andrea Trescot, MD)

추가 내용

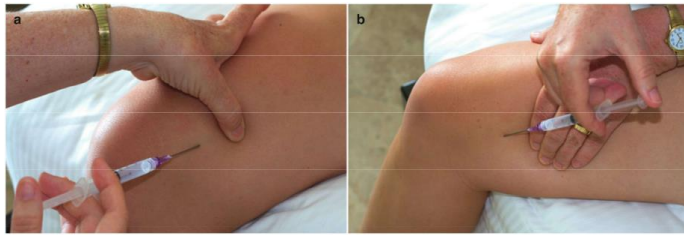


Fig. 58.16 Injection of the saphenous nerve in the adductor canal. (a) Distal to proximal; (b) proximal to distal (Image courtesy of Andrea Trescot, MD)

• Adductor canal approach

- 블라인드 시술법으로 양와위로 무릎을 약간 굽힌 상태로 누워서 허벅지 안쪽 중앙에서 내측광근과 봉공근 사이의 평면을 찾고, 그 사이를 치료 포인트로 잡는다.

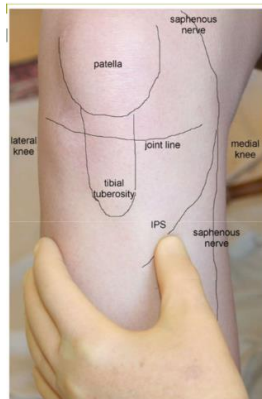
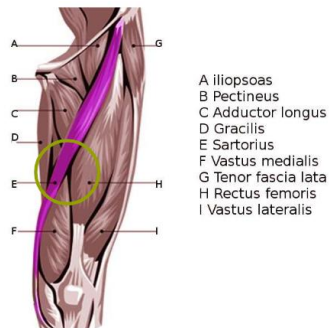


Fig. 58.13 Physical exam of the infrapatellar saphenous nerve (Image courtesy of Andrea Trescot, MD)

(근위부)

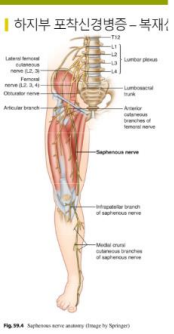
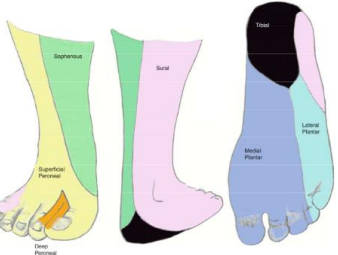
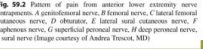


Fig. 58.17 Landmark-guided injection of the infrapatellar saphenous nerve (Image courtesy of Andrea Trescot, MD)

- Injection of the IPS is also performed with the knee flexed. The non-injecting hand identifies the point of maximum tenderness, with the index and middle fingers straddling the nerve from below. The needle enters from below and is directed toward the tibial tubercle and advanced to bone.

- 혈해혈의 약간 위쪽 부위로 접근할 수도 있다.

3. 복재신경 포착신경병증(원위부)

증상	발목 내과 전상방-발내측 통증, 감각이상
원인	발목 수술 꼭끼는 신발
포착 위치	정강이 원위부 내측 내과 부위
통증 부위	- 여러 신경 분절이 지배하는 피부영역이 다르기 때문에 통증 부위를 통해 치료 포인트를 정할 수 있다.  Fig. 99.4 Saphenous nerve anatomy (Image by Springer)  Fig. 99.3 Distribution of the nerves of the foot (Image courtesy of Michael Brown, MD)  Fig. 99.2 Pattern of pain from anterior lower extremity nerve entrapments. A peroneal nerve, B femoral nerve, C lateral femoral cutaneous nerve, D obturator, E lateral tibial cutaneous nerve, F saphenous nerve, G superficial peroneal nerve, H deep peroneal nerve, I sural nerve (Image courtesy of Andrea Trescot, MD) Fig. 99.1 Patient's pain distribution map (Image courtesy of Michael Brown, MD)  - 발목 내측의 통증을 호소하는 경우가 많다.

영상

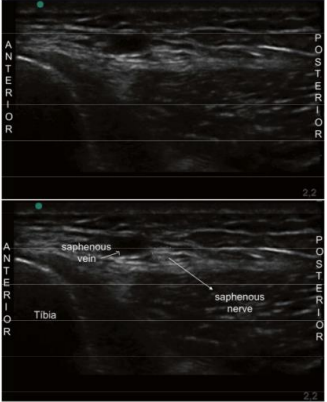
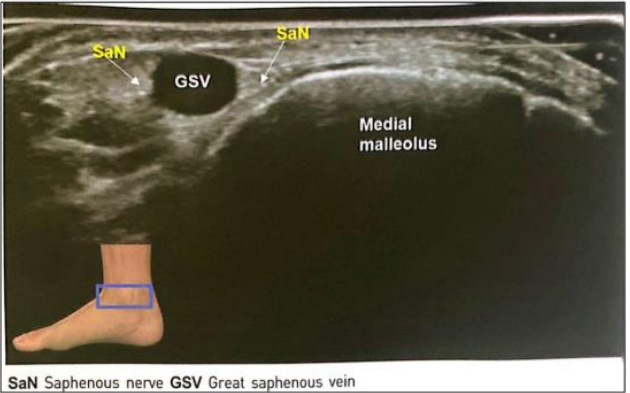


Fig. 99.11 Ultrasound image of the distal saphenous nerve at the tibia, anteroposterior and lateral (Image courtesy of Thiago Neuser Probst, MD)



Fig. 99.10 Ultrasound probe on the distal saphenous nerve at the distal tibia (Image courtesy of Andrea Trescot, MD)

- 초음파 검사를 통해 내과점 부위에서 복재신경이 나뉘는 것을 확인할 수 있다.



SaN Saphenous nerve GSV Great saphenous vein

추가 내용



Fig. 59.9 Landmark-guided distal saphenous nerve injection (Image courtesy of Andrea Trescot, MD)

- The needle should be placed parallel to the nerve (Video 59.2) (Fig. 59.9), and a peripheral nerve stimulator may be useful. For diagnostic purposes, infiltration of 1 cc of local anesthetic at the point of maximum tenderness is used.

- 블라인드 시술법으로 바늘은 신경과 평행하게 들어가야 한다.
(앞의 복재신경 근위부의 내용과 겹치는 내용이 많아서 금방 넘어가셨습니다.)